ECE 554 Vocoder Project Proposal

1. Team Introduction

The team is composed of the following members: Kalyan Sriram, Avery Dudra, Jonathan Wang, and Vishrut Agrawal. Kalyan's interests lie in computer architecture and compiler design. Avery's interests lie in computer architecture. Jonathan's interests lie in controls. Vishrut's interests lie in signals and systems.

2. Project Overview

The goal of this project is to implement a real-time autotuning system that captures live vocal input from a microphone, detects the current pitch using autocorrelation, and corrects it to match a user-selected target pitch provided through a keyboard interface.In addition, the system will include a real-time graphics component that displays relevant audio or system data, such as detected pitch, target pitch, or waveform information.

Real-time pitch correction is widely used in music production and vocal training to help singers maintain accurate pitch and improve overall sound quality. However, most autotuning systems rely on software running on general-purpose processors, which can introduce latency and limit real-time responsiveness. Implementing pitch detection and correction on an FPGA enables low-latency, deterministic processing, making it suitable for live applications where immediate feedback is essential.

3. Technical Approach

The system will be implemented as a real-time audio processing pipeline on the DE1-SoC FPGA using the onboard audio codec for microphone input and speaker output via the I2S interface. Pitch detection will be performed using an autocorrelation-based algorithm, which computes the fundamental pitch period of the input signal over a sliding window. Keyboard input will be captured through FPGA GPIO and converted into a target pitch. Pitch correction will be implemented using the PSOLA algorithm, where the input waveform is segmented based on detected pitch periods and reconstructed using overlap-add techniques to produce a pitch-shifted output. The corrected audio will be transmitted back to the codec for playback. In addition, the FPGA will transmit pitch or system data to a Raspberry Pi, allowing the generation of real-time graphical visualizations of the system's operation.

This project is well suited for ECE 554 because for a real-time autotuning system, FPGA acceleration is essential because both autocorrelation and PSOLA involve computationally intensive real-time signal processing and continuous sample-by-sample computations.

4. Execution Strategy

Anticipated Challenges:

1) Autocorrelation module operates efficiently in real time without introducing excessive latency or requiring excessive hardware resources
2) Managing memory buffering and timing for PSOLA, since waveform segments must be stored, repositioned, and overlap-added without disrupting the continuous audio stream
3) Maintaining smooth transitions between pitch segments in PSOLA, as incorrect alignment or timing can introduce audible artifacts such as clicks, distortion, or discontinuities

Risk Management:

1) The autocorrelation module will first be tested using known test signals (e.g., sine waves) to verify correctness before applying it to live vocal input
2) Memory buffering for PSOLA will be carefully designed using FPGA block RAM, and buffer sizes will be adjusted as needed to ensure continuous real-time operation
3) If full PSOLA implementation proves too complex within the project timeline, a simplified pitch shifting method using controlled overlap-add with fixed timing will be used as a fallback to demonstrate functional pitch correction

5. Project Timeline & Evaluation

Midterm Success Criteria:

1) Working audio passthrough system (microphone input to speakers)
2) Working keyboard input and correct conversion of key presses to pitch target
3) Functional autocorrelation pitch detection module that can detect and output pitch of simple test signals (sine waves, sustained vocal input).

Final Success Criteria:

1) Fully integrated real-time autotuning system that captures live microphone input, detects pitch using autocorrelation, and corrects it using PSOLA.
2) System successfully adjusts vocal pitch to match the selected keyboard target pitch in real time.
3) Smooth pitch correction without significant audio glitches, distortion, or interruptions..

Task	Week 1 (2/23-2/27)	Week 2 (3/2-3/6)	Week 3 (3/9-3/13)	Week 4 (3/16-3/20)	Week 5 (3/23-3/27)	Week 6 (3/30-4/3)	Week 7 (4/6-4/10)	Week 8 (4/13-4/17)	Week 9 (4/20-4/24)
Project Planning									
Codec configuration (I2C module)									
Mic input and Speaker output (I2S transmitter and Receiver)									
Keyboard input integration (MIDI Reciever Module)									
Audio passthrough testing									
SW implementation of autocorrelation									
SW implementation of PSOLA						Spring Break			
HW implementation of autocorrelation									
HW implementation of PSOLA									
Full System Integration (audio + keyboard + pitching)									
Real-time Testing and Debugging									
Raspberry Pi setup & Communication Interface									
FPGA to RaspPi data transfer implementation									
Graphics visualization development									
Final Real Time Testing									
Additonal Weekly Notes:	Lightning Talk 2/26	Architectural Review 3/4	Micrarchitectural Review 3/12		Module Implementation Demo 3/24				Internal Demo 4/23

6. Team Responsibilities

System Integration (All)
Pitch Detection (Vish and Jonny)
Pitch Correction (Avery and Kalyan)
Visual Integration (All)